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ORIGINAL PAPER

An Insight into the Mechanical and Fracture Characterization of Minerals and Mineral Interfaces in Granite Using Nanoindentation and Micro X-Ray Computed Tomography

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Abstract

It is extremely important to investigate the mechanical properties and failure characteristics of granite at the mesoscale to understand the mesoscopic evolution mechanism of the time-dependent fracture of deep-buried hard rocks. This study explores the mesoscopic mechanical properties (i.e., hardness, Young's modulus, and fracture toughness) of the primary granite minerals and their interfaces using nanoindentation. Micro X-ray computed tomography was used to analyze the fracture characteristics of the failed Brazilian disc of granite. Moreover, two homogenization upscaling methods were used to calculate granite's Young's modulus and compared with that from uniaxial compression test results. The results demonstrate the following: (1) The mechanical properties of granite minerals are related to the peak load of nanoindentation. Young's modulus and hardness of quartz, K-feldspar, and plagioclase decrease with an increase in the peak load and tend to be stable when the peak load reaches 5000 μN, whereas Young's modulus and hardness of biotite at multiple peak loads are in variability and irregularity. (2) Young's modulus and hardness of quartz–plagioclase and quartz–biotite interfaces are between quartz and plagioclase and quartz and biotite, respectively. Furthermore, a linear formula is proposed to estimate fracture toughness based on Young's modulus of granite minerals. (3) The generalized means method that predicts granite's Young's modulus is more accurate compared to the Mori–Tanaka method. (4) Under Brazilian splitting conditions, ~ 84% of tensile cracks along intragranular cracks primarily occur in feldspar minerals, while ~ 16% along grain boundaries occur at feldspar–biotite interfaces.

Highlights

- Mesomechanical properties of granite minerals and mineral interfaces.
- A linear relationship between the fracture toughness and Young's modulus.
- Homogenization methods used for upscaling the mesoscopic Young's modulus of granite.

Keywords Deep-buried granite · Modulus · Mineral interface · Fracture toughness · Upscaling analysis · Nanoindentation · Micro X-ray computed tomography

List of Symbols

A_c	Contact area of the indenter on the sample surface
A_{max}	Maximum contact area of the indenter on the sample surface
E and E_r	Young's modulus and reduced modulus of the sample
E_{hom}	Homogenization Young's modulus of granite

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E_i	Young's modulus of the indenter
E_u	Young's modulus of the weakest mineral
E_w	Uniaxial compressive modulus of granite
f_i	Volume fraction of minerals
G_c	Critical energy release rate
G_{hom}	Shear modulus of granite
H	Hardness of the sample
h	Indentation depth of the sample
h, h_c, h_f, and h_m	Indentation depth, contact depth, residual depth, and maximum indentation depth
J	Scaling parameter
K_{hom}	Bulk modulus of granite
K_{IC}	Tensile (mode I) fracture toughness
k_s and k_{low}	Bulk moduli of the biotite and biotite porous matrix
l	Crack length
M	Specific mechanical property
P	Applied loading force of the indenter
P_{max}	Peak force of the indenter
S	Contact stiffness of the sample
U_{crack}, U_{pp}, U_{ir}, U_t, and U_e	Fracture energy, pure plasticity energy, irreversible energy, total energy, and recoverable elastic energy
V	Volume fraction of the weakest mineral
α and β	Power law fitting constants
γ_p	Plastic energy
ω	Porosity of the biotite pore matrix
ε	A constant that depends on the indenter tip
μ_s and μ_{low}	Shear moduli of the biotite and biotite porous matrix
ν and ν_i	Poisson's ratios of the sample and indenter
σ_f	Critical fracture stress
E_m, c and φ	Young's modulus, cohesion, and internal friction angle of minerals and mineral interfaces

1 Introduction

Deep-buried rock mass has considerably different mechanical properties compared to shallow-buried rock mass because of the high geostress environment. An obvious difference is the progressive fracture of rock mass induced by high geostress (Jiang et al. 2021; Xie et al. 2021). Engineering practice demonstrates that even hard and intact deep-buried granites produce obvious time-dependent fractures under excavation unloading, decreasing the strength of the rock mass over time and leading to the failure of the supporting structure under excessive load, which result in the instability of the surrounding rock. The deformation and failure characteristics of deep-buried hard rock are not only affected by its geological environment but also closely related to the crystal structure of minerals in the rock (Yang et al. 2020). The generation, development, and penetration of microcracks in mineral crystals cause the macroscopic fracture/failure of rock. Researchers must understand the mechanical properties of the mineralogical components at the mesoscale because of the diversity of mineralogical components in crystalline rocks and their complex mechanical properties (Ma et al. 2021; Maruvanchery and Kim 2020). Therefore, it is valuable to examine the mechanical properties of mineralogical components in granite to understand the deformation and failure mechanisms of the deep-buried rock.

Traditionally, researchers determined the deformation and failure characteristics of rocks using laboratory uniaxial/triaxial compression testing of cylindrical rock samples. They then combined this with in situ observation results of rock mass failure to extrapolate the deformation and failure mechanisms of the engineering rock mass. Given the rapid development of nanoexperimental mechanics, nanoindentation is popular in the research field of alloy, ceramics, and composite materials because of its accurate displacement and force sensitivity. Furthermore, the use of nanoindentation is increasingly expanding in the field of rock mechanics (Constantinides et al. 2006; Ma et al. 2020). Recently, researchers have extensively used statistical nanoindentation to examine the mechanical properties of rock minerals. In these tests, several grid nanoindentations are performed in a selected area, and the test data are then deconvoluted using the Gaussian mixture model to obtain the mechanical parameters of the primary minerals and determine the probability distribution of mechanical parameters (Akono et al. 2019; Luo et al. 2021; Miller et al. 2008; Sobhbidari and Hu 2021; Wu et al. 2020; Xu et al. 2020). Dynamic stiffness can be measured using continuous stiffness measurement or frequency-specific dynamic indentation mode in nanoindentation. The continuous functions of the hardness and Young's modulus with indentation depth can be obtained in a single indentation test, which has been used to

examine the mesomechanical properties of shale, concrete, and other materials (Li et al. 2021; Yang et al. 2020). Most recently, atomic force microscopy (AFM) has been applied to obtain surface adhesion, Young's modulus, deformation, and energy dissipation of the material (Chen et al. 2021; Eliyahu et al. 2015; Li et al. 2020; Sobhbidari and Hu 2021). Compared with traditional laboratory tests, the aforementioned method not only has no limitations on the size and shape of the sample but also saves time and money.

Multiple nanoindentations have been performed to understand the mesomechanical behavior of certain porous rocks (e.g., sandstone, shale, and coal) (Akono et al. 2019; Li et al. 2020, 2021; Liu et al. 2016; Liu and Ostadhassan 2017; Manjunath and Jha 2019; Sun et al. 2021; Wu et al. 2020; Yang et al. 2020). However, only a few data are available on the mesomechanical properties of individual minerals and mineral interfaces of deep-buried hard rocks such as granite (Ma et al. 2021; Maruvanchery and Kim 2020). This study explores the mesoscale characterization of a deep-buried granite from the Shuangjiangkou Hydropower underground powerhouse. Nanoindentation based on the phase position of scanning electron microscopy–electron-dispersive spectroscopy (SEM–EDS) is used to examine the relationship between different peak loads and the mesomechanical parameters of the primary minerals and mineral interfaces of granite. The mesomechanical properties, including hardness, Young's modulus, and fracture toughness of each mineral and mineral interface are analyzed. Micro X-ray computed tomography (CT) was used to examine the fracture characteristics of the failed Brazilian disc and the proportion of two fracture modes (intragranular and intergranular crack). Finally, two homogenization upscaling methods are used to calculate granite's Young's modulus and compared with that of uniaxial compression results.

2 Materials and Methods

2.1 Sample Preparation

The granite used for preparing experimental samples was obtained from the Shuangjiangkou Hydropower underground powerhouse in southwestern China, with a buried depth of 320–500 m and a maximum principal stress of 37.82 MPa. The results from polarizing microscopy measurements (Table 1) show that the volumetric fractions of the main minerals in the Shuangjiangkou granite, i.e., quartz, plagioclase, biotite, K-feldspar, and chlorite, are 40%, 40%, 10%, 7%, and 3%, respectively. The grain sizes of these minerals are 300–4500, 350–3000, and 200–1500 μm, respectively.

The nanoindentation testing samples were prepared as follows: First, granite was cut into discs having a thickness of 1 cm, where the top and bottom surfaces are paralleled, and then divided into four sections for subsequent polishing. Second, the samples were polished using sandpaper with different grain fineness levels of P800, P1200, P2500, P5000, and P7000 using a Buehler polishing instrument. Third, the samples were polished with ultrafine alumina suspension to eliminate surface roughness as much as possible. Finally, the samples were placed in a 45 °C environment and dried for 24 h. For the entire polishing process, the samples were cleaned with anhydrous ethanol to ensure flat, clean, and smooth sample surfaces (Fig. 1a).

2.2 Mineral-Positioned Nanoindentation

2.2.1 Mineral Position

The primary mineral compositions were identified by scanning the polished smooth granite surface with SEM–EDS (Fig. 2). The type of each mineral is determined using the element distribution in SEM–EDS diagrams and the representative elements of each mineral in granite. Figure 2b–f shows the element distributions of Si, Al, K, Ca, and Na. The region with the highest Si enrichment without Al represents quartz. Al is unique to K-feldspar and plagioclase, and its enrichment area represents these two minerals. The plagioclase can be determined based on the distribution of Na and Ca elements. Then, the distribution area of K-feldspar can be determined. The mineral types at different positions on the granite surface can be identified by analyzing the distribution characteristics of the above-mentioned elements.

Table 1 Mineral composition of the granite sample

Mineral	Characteristic chemical formula	Grain size (μm)	Volumetric fraction (%)
Quartz	SiO_2	300–4500	40
K-feldspar	$KAlSi_3O_8$	350–3000	7
Plagioclase	$NaAlSi_3O_8$–$CaAl_2Si_2O_8$	350–3000	40
Biotite	$K(Mg, Fe^{3+})_3AlSi_3O_{10}(F, OH)_2$	200–1500	10
Chlorite	N/A	N/A	3

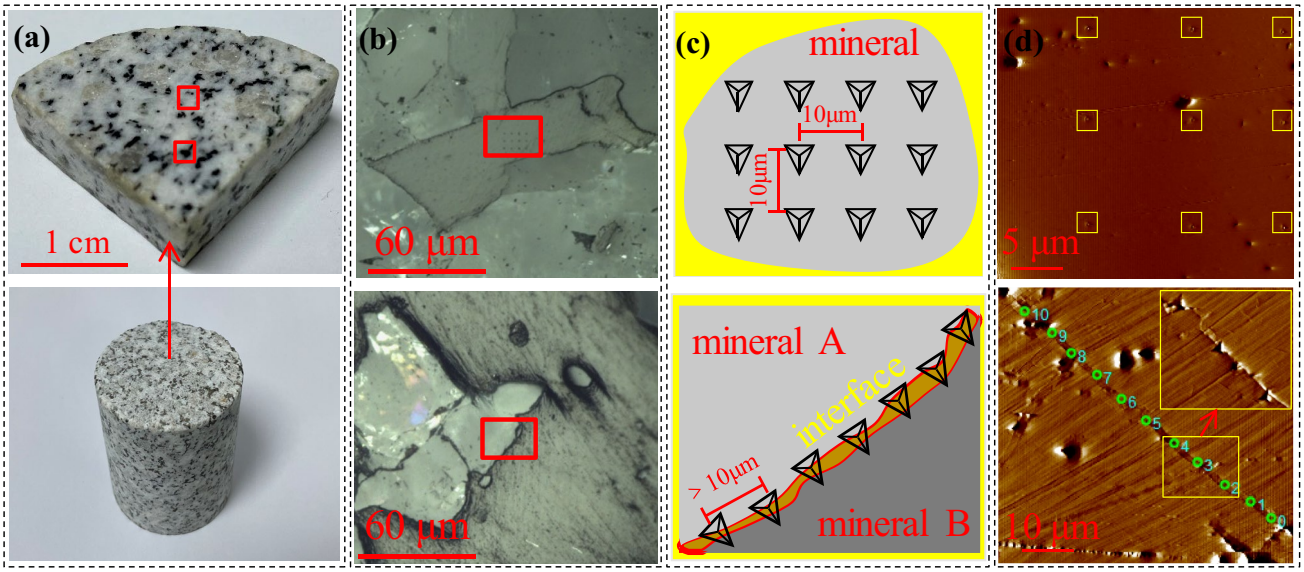


Fig. 1 Nanoindentation testing: **a** the polished test sample; **b** high-speed performance imaging of test areas; **c** schematic showing nanoindentation locations across the prepared sample; **d** residual indentation morphology of the mineral and interface

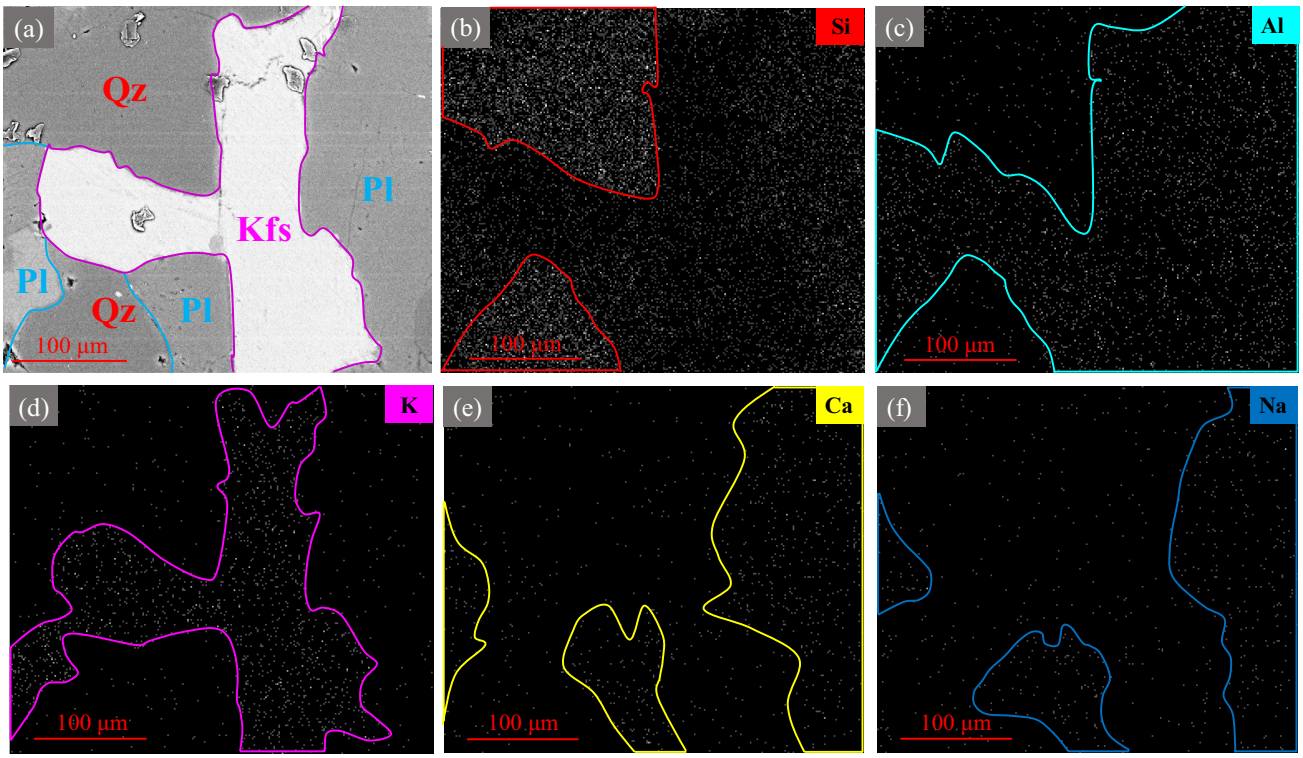


Fig. 2 SEM micrographs and EDS chemical element mappings of polished samples: **a** backscattered electron image of the target region; **b** region with the highest Si enrichment without Al, indicating quartz; **c** region with the highest Al enrichment, indicating K-feldspar and plagioclase; **d** region with the highest K enrichment, indicating K-feldspar; **e** and **f** regions with the highest Ca and Na enrichment, indicating plagioclase

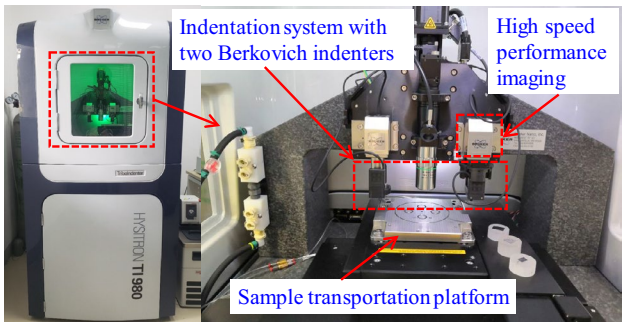


Fig. 3 Hysitron Ti 980 Triboindenter apparatus illustrating the major components

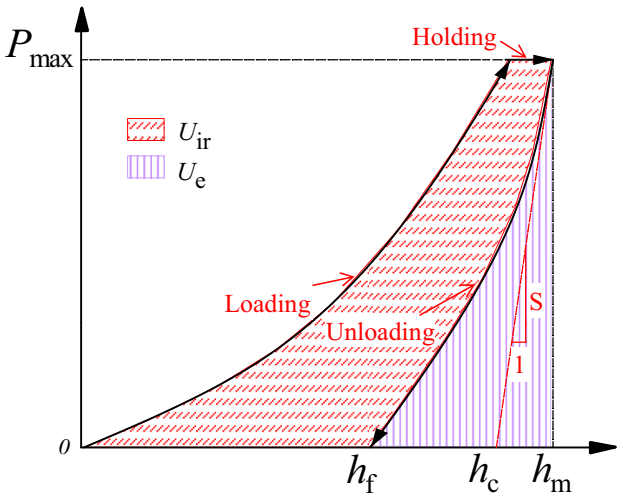


Fig. 4 Schematic illustration of the indentation load–depth curve and energy analysis for calculation

2.2.2 Nanoindentation Testing

To examine the relationship between the peak load of nanoindentation and the mesomechanical parameters of the minerals, nanoindentations with peak loads of 2000, 3000, 4000, 5000, 6000, 7000, and 8000 μN were performed on quartz, K-feldspar, and plagioclase. An excessive peak load may destroy biotite because of its soft texture, thus affecting the accuracy of the test results. Therefore, the peak gradient moderately decreases, and the peak loads are 2000, 3000, 4000, 5000, and 6000 μN. In this study, the Hysitron Ti 980 Triboindenter apparatus equipped with two Berkovich indenters was used for nanoindentation testing (Fig. 3). During the experiment, a high-speed performance imaging probe was attached to the Triboindenter apparatus. It facilitated the identification of the mineral area confirmed by SEM–EDS (Fig. 1b) as the indentation array area. The indentation array (3 × 4) spacing was set to 10 μm (Fig. 1c), thus satisfying the requirement of greater than 20 times the maximum indentation depth (Miller et al. 2008). Mineral indentation tests were then conducted at the indentation point under different peak loads. The loading mode of 10 s (loading)–5 s (holding)–10 s (unloading) was used, with the loading and unloading rates being constant in the loading and unloading stage. The 5s holding period is primarily used to eliminate strain lag during the loading process (Miller et al. 2008). The indentation morphology was obtained after the test by scanning the test area with the indenter (Fig. 1d). For the mineral interfaces, ten indentations were performed along with the interfaces (Fig. 1c). The locations of indentation on each mineral interface were selected based on the chemical composition analysis of minerals by EDS and from observations made using an optical microscopy. The implementation of the mineral interface nanoindentation shows quartz–K-feldspar, quartz–plagioclase, quartz–biotite, K-feldspar–plagioclase, and K-feldspar–biotite interfaces. The indentation of the plagioclase–biotite interface was not performed, because it was not present in the test.

Nanoindentation technology uses diamond tips with known geometric shapes (such as the triangular pyramid and spherical shapes) and mechanical parameters to press in the sample surface with minimal load. During the test, the applied load P continuously increases, thus resulting in an increase in depth h. Typical experiments are performed in the following ways: increasing the load continuously to reach the defined value P_{max}, holding it for a few seconds, and then unloading it continuously (Fig. 4).

The initial unloading slope for linear isotropic elastic materials can be achieved by the elastic solution related to the applied loading force P and the indentation depth h. The initial unloading slope is known as contact stiffness S (Oliver and Pharr 1992), which is calculated by analytically differentiating Eq. (1) at the maximum depth.

To describe the nonlinear load–depth curve of the unloading segment, the Oliver–Pharr method (Oliver and Pharr 1992) considers fitting the unloading curve of the residual depth h_f to an exponential form, $P = \alpha(h - h_f)^{\beta}$, and then analytically differentiating this expression at the maximum indentation depth h_{max} to obtain the elastic contact stiffness S as follows:

$$S = \left(\frac{dP}{dh}\middle| h = h_{max}\right) = \alpha\beta(h_{max} - h_f)^{\beta-1}, \quad (1)$$

where α and β are the power law fitting constants.

The contact depth h_c and the projected contact area of the indenter on sample surfaces A_c can be calculated using the following two equations:

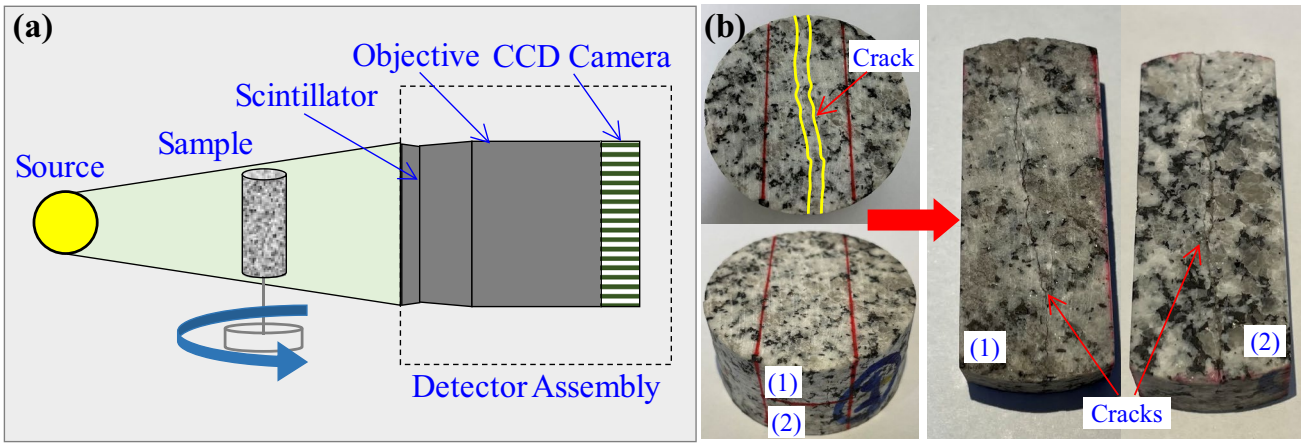


Fig. 5 Micro CT testing: **a** schematic of μCT; **b** the failed Brazilian disc sample

$$h_c = h_{max} - \varepsilon \frac{P_{max}}{S} \tag{2}$$

$$A_c = 24.56 h_c^2, \tag{3}$$

where ε is a constant that depends on the indenter tip and is set as 0.75 for the Berkovich indenter and P_{max} is the maximum indentation load.

Once the contact depth h_c and contact area A_c are determined, the indentation hardness H and reduced modulus E_r can be calculated using the following two equations:

$$H = \frac{P_{max}}{A_c} \tag{4}$$

$$E_r = \frac{\sqrt{\pi}}{2\beta} \frac{S}{\sqrt{A_c}}. \tag{5}$$

The relationship between Young's modulus and reduced modulus of the sample is determined as follows:

$$E = (1 - \nu^2)\left(\frac{1}{E_r} - \frac{1 - \nu_i^2}{E_i}\right)^{-1}, \tag{6}$$

where E_i, E, ν_i, and ν are Young's moduli and Poisson's ratios of the indenter and sample, respectively. For the diamond indenter, $E_i = 1140$ GPa and $\nu_i = 0.07$. The Poisson ratio of the sample can be reasonably assumed as per the relevant literature (Mavko et al. 2009; Zhang et al. 2017). The sensitivity analysis of 40% uncertainty of Poisson's ratio demonstrates that the uncertainty of Young's modulus is only 5% (Hay 2009).

2.3 Micro X-Ray CT of a Brazilian Disc

This study uses a Brazilian disc test to determine the tensile strength of the rock samples and micro X-ray CT to demonstrate the heterogeneity of the internal mesostructure of rock samples. Multiple 2D X-ray projections obtained in CT are used to reconstruct the 3D internal structures of the samples containing the failure path. Based on the grain size, the crack patterns within the samples and their distribution in granite minerals are counted to characterize the damage and failure behavior of samples.

The Zeiss Xradia Versa 410 μCT system was used for the μCT scanning of samples after the Brazilian disc test. A total of 2140 images having a sensor at 0.17° rotation increments were obtained per scan (Fig. 5a). To improve the imaging quality and account for the similar attenuation characteristics because of the similar density of quartz and feldspar, the sample before the μCT test is processed, as shown in Fig. 5b. Moreover, the sample thickness is reduced by half, thereby improving the X-ray penetration ability and ensuring sufficient resolution and phase contrast to differentiate each mineral (Mahabadi 2012).

3 Results and Discussion

3.1 Load–Indentation Depth Curves

The load–indentation depth curve is the basis for analyzing the deformation behavior and mechanical properties (E and H) of individual minerals and mineral interfaces in rock samples at the mesoscale. Three load–indentation depth curves for minerals and mineral interfaces were selected at a peak load of 5000 μN (Fig. 6a, b).

The coincidence degree of quartz from the load–indentation depth hysteresis curves is high, and the curves are

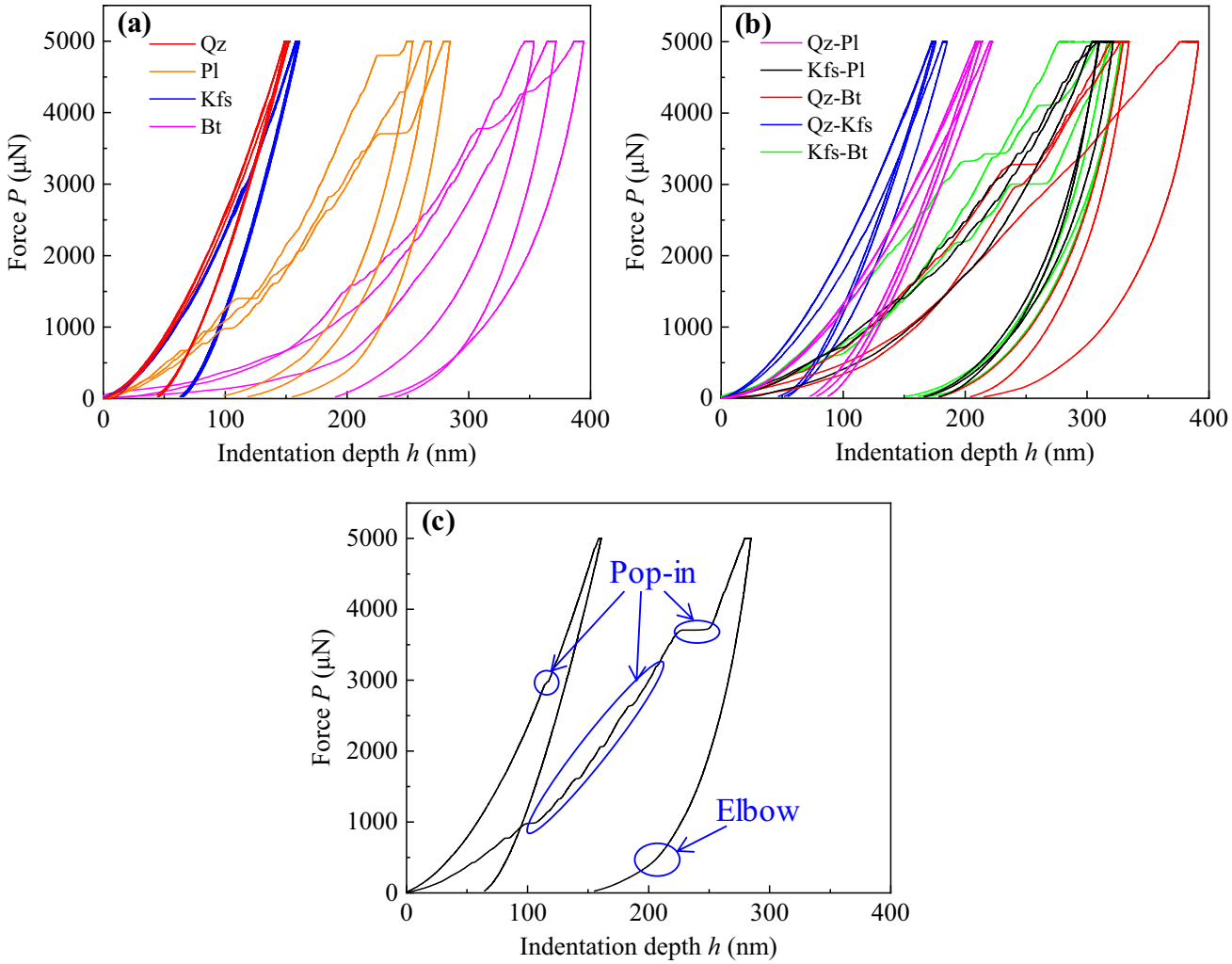


Fig. 6 Load–indentation depth curves for different granite minerals and interfaces: **a** minerals; **b** interfaces; **c** "pop-in" and "elbow" phenomena in nanoindentation curves

smooth in the loading and unloading stages. There is almost no creep phenomenon in the holding stage. The maximum indentation depth is not > 150 nm, and the elastic recovery in the unloading stage is large, thus indicating that the quartz has good homogeneity, hard texture, high hardness, and high Young's modulus. Note that these characteristics of the curves of K-feldspar, quartz–K-feldspar, and quartz–plagioclase interfaces are similar to those of quartz. The indentation depth of biotite sharply increases in the early loading stages, which is considered to be compacted along the normal cleavage direction. With increase in load, the rate of indentation depth slowly increases. The important creep phenomenon of biotite occurs in the holding stage. The elastic deformations recover after unloading, thus leaving the maximum unrecoverable plastic deformations. Similar to biotite, the heterogeneity of plagioclase can be represented by the lower coincidence and smoothness of curves. The curve positions of K-feldspar–plagioclase, K-feldspar–biotite, and quartz–biotite are dispersed, indicating that their mechanical properties are variable. The curves become concave and convex as the load increases, which may be attributed to the material holes or the heterogeneity of the surface layers of samples in the indentation process (Tyurin et al. 2016). During the holding stage, the creep at mineral interfaces is more pronounced than that at minerals.

Figure 6c shows the "pop-in" phenomenon in the loading curve where the indentation depth sharply increases with a minor change in load (Zhang et al. 2009). This is attributed to the heterogeneity of the contact material and the fact that as the load increases, the elastic energy of the material in the contact area gradually increases, thus resulting in a plasticity initiation (elastic-to-plastic transition). Consequently, the first "pop-in" occurs (Liu and Ostadhassan 2017; Luo et al. 2020; Pöhl 2019). Certain unloading curves follow the "elbow" pattern. This is because during the unloading process, the pressure of the indenter is released, the material

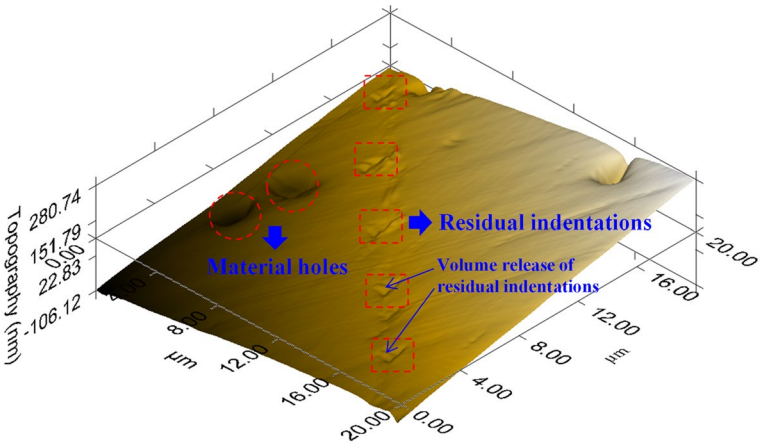


Fig. 7 Topographic scanning map of residual indentations

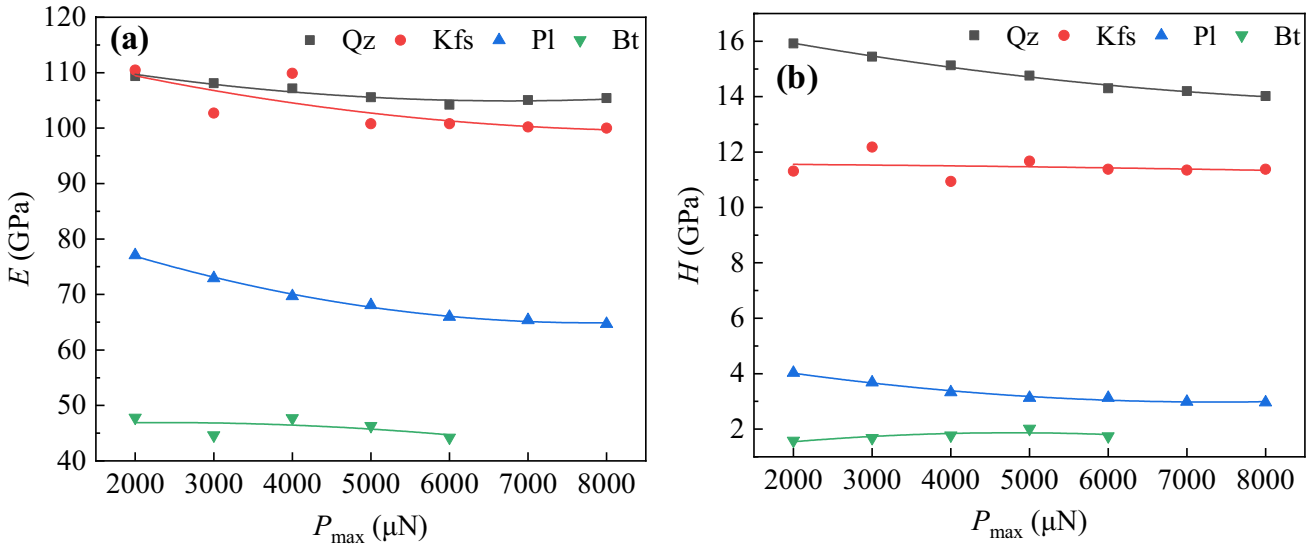


Fig. 8 Mechanical parameters of minerals under different peak loads: **a** Young's moduli of different minerals; **b** hardness of different minerals

is slowly transformed from crystalline to amorphous phase, and the volume of the amorphous phase expands, thus contributing to the indenter uplift (Domnich et al. 2000; Liu et al. 2016). Figure 7 shows the volume release in the residual indentation. Once the pressure exceeds the critical point, the phase transformation occurs, i.e., from elastic to plastic and from crystalline to amorphous (Juliano et al. 2003; Tabor 1978). All indentations having distinct "pop-in" or "elbow" phenomena primarily occur in plagioclase and biotite and mineral interfaces containing either of them (Fig. 6a, b).

3.2 Young's Modulus and Hardness

The relationship between the mesomechanical properties of the primary minerals and peak load (Fig. 8) shows that the Young's moduli and hardness of quartz, K-feldspar, and plagioclase decrease as the peak load increases and tend to stabilize after 5000 μN. However, Young's modulus and hardness of biotite remain in a relatively stable range; therefore, the following discussion is based on the mechanical parameters at a peak load of 5000 μN.

The Oliver–Pharr method is then used to calculate Young's modulus and hardness of minerals and mineral interfaces based on the testing data shown in Fig. 9. Quartz, K-feldspar, and Qz–Kfs interface demonstrate the highest Young's modulus and hardness of 106.3, 100.6, and 91.6 GPa and 14.5, 11.4, and 12.1 GPa, respectively. The Young's modulus and hardness of plagioclase and biotite, which are weak-phase minerals in granite, are 67.2 and 45.2 GPa and 3.2 and 1.8 GPa, respectively. The mechanical properties of the mineral interface between strong-phase minerals (i.e., quartz and K-feldspar) and weak-phase minerals (i.e., plagioclase and biotite) are almost between strong- and weak-phase minerals, except for the Kfs–Pl interface. The normal distribution curves

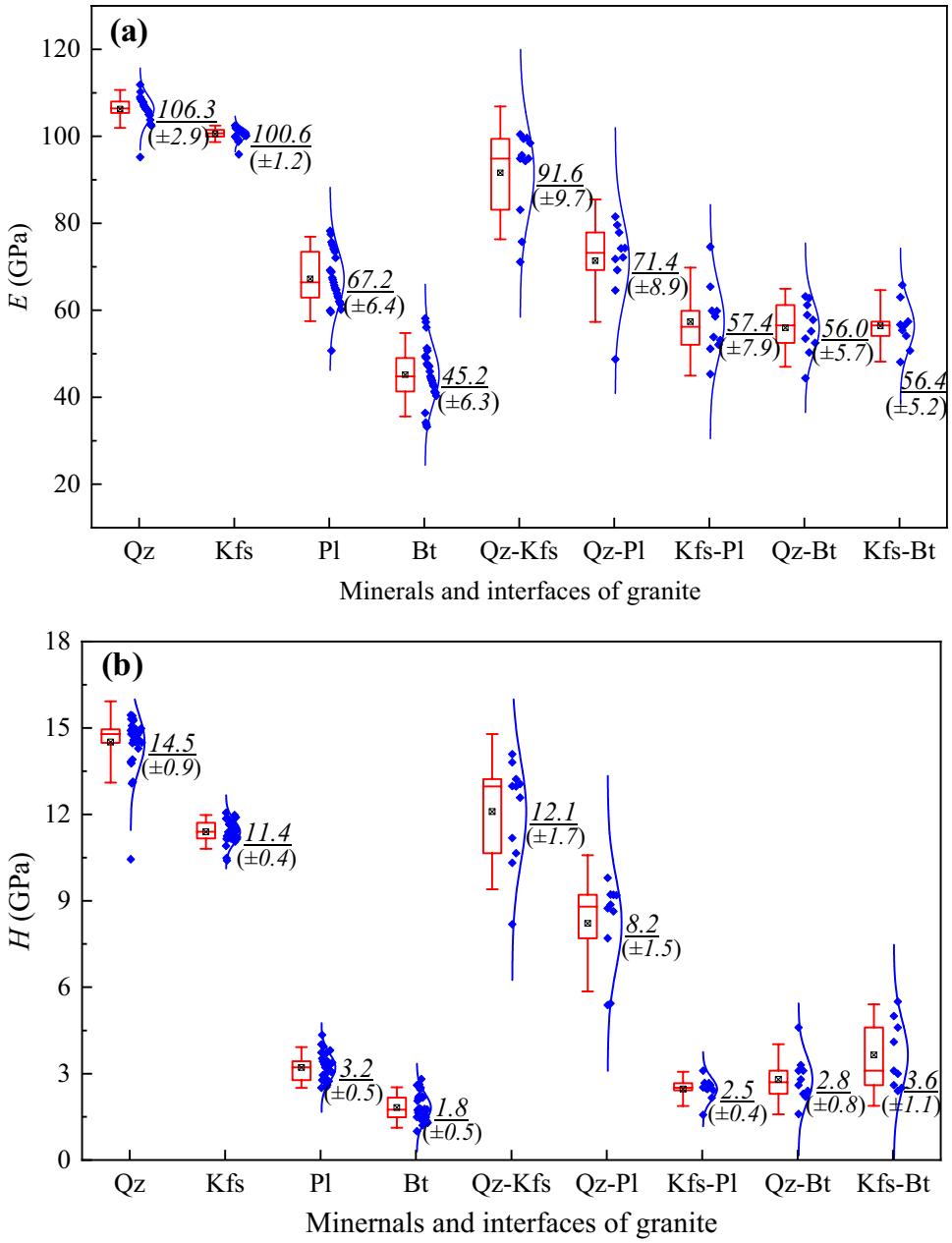


Fig. 9 Mechanical properties of minerals and mineral interfaces in granite: **a** Young's modulus distribution; **b** hardness distribution. The symbol "⊠" on each box indicates the mean value, the central mark indicates the median, and the bottom and top edges of the box indicate the 25th and 75th percentiles, respectively; the blue curve is the normal distribution curve of the testing data

indicate that the variability of the mineral interface testing data is greater than that of the mineral, which could indicate that the density of flaws and heterogeneity of the mesostructure at the mineral interface are greater than those of the mineral.

Nanoindentation is a multiphysics process that involves elastic deformation, plastic deformation, and brittle failure. The material properties control the physical process transformation (Manjunath et al. 2019). Statistical data demonstrate a positive correlation between the hardness and Young's modulus of minerals and mineral interfaces, thus indicating that materials with strong resistance to plastic deformation often have strong resistance to elastic deformation (Fig. 10). Moreover, the goodness of fit, R^2, of the hardness and Young's modulus data of minerals are greater than that of mineral interfaces, which is consistent with the variability of testing data depicted by normal curves (Fig. 9).

3.3 Fracture Toughness

Fracture toughness characterizes the ability of materials to prevent crack propagation and is a quantitative index for measuring material toughness (Shi et al. 2019). Most prior indentation-based studies are limited to the framework of elastic properties using the Oliver–Pharr method, whereas

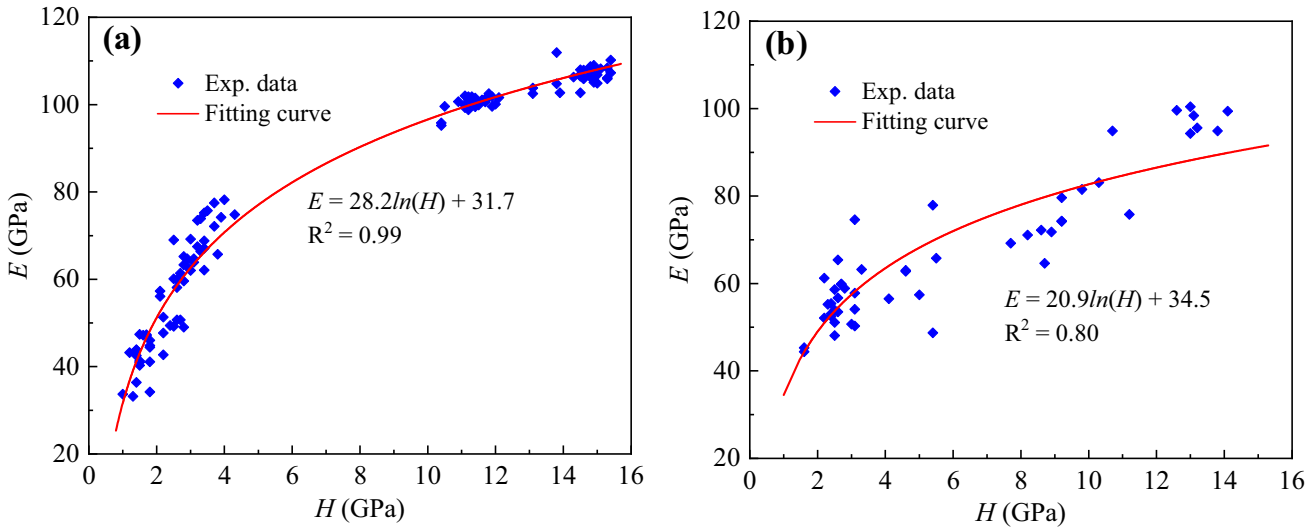


Fig. 10 Relationship between the hardness and Young's modulus of minerals and mineral interfaces in granite determined by nanoindentation testing: **a** minerals; **b** mineral interfaces

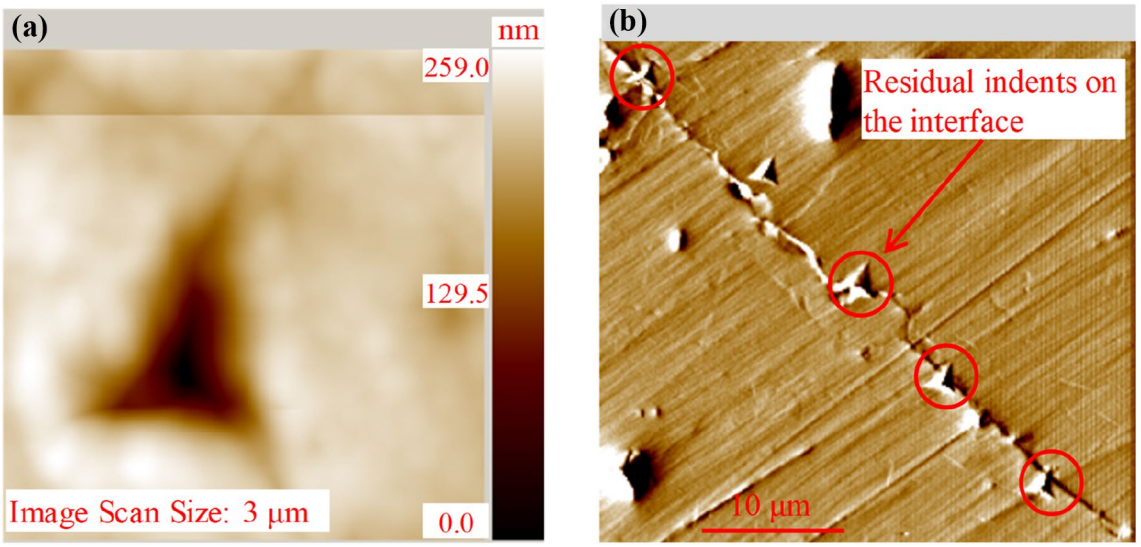


Fig. 11 Residual indentation morphology scanned by the indenter: **a** depth mapping of a residual indent on a biotite particle; **b** residual indents on the Qz–Kfs interface

there are limited studies on the evaluation of their fracture properties (Liu et al. 2016). In this study, a nanoindentation technique was used to examine the fracture toughness of granite samples. As shown in Fig. 11, the granite sample had almost no surface cracks. Moreover, it is difficult to measure the crack length at such a small scale. Consequently, this study adopted the energy analysis method (Liu et al. 2016; Liu and Ostadhassan 2017; Sun et al. 2020) to obtain the fracture toughness of granite minerals and mineral interfaces.

The theoretical concept behind the energy analysis method assumes that the fracture energy U_{crack} can be calculated from the relationship between pure plasticity energy U_{pp} and irreversible energy U_{ir} (Cheng et al. 2002) and expressed as Eq. (7)

$$U_{crack} = U_{ir} - U_{pp} = U_t - U_e - U_{pp}, \tag{7}$$

where the total energy U_t, irreversible energy U_{ir}, and recoverable elastic energy U_e can be represented and calculated by the areas in Fig. 4. The energy caused by the pure plasticity

Table 2 Fracture toughness of minerals, mineral interfaces, and rocks

Fracture toughness (MPa·m$^{0.5}$)	Value	Testing method
Mineral		Nanoindentation
Qz	5.34 (±0.07)	
Kfs	4.83 (±0.16)	
Pl	2.70 (±0.30)	
Bt	2.41 (±0.34)	
Mineral interface		
Qz–Kfs	4.96 (±0.50)	
Qz–Pl	4.00 (±0.76)	
Kfs–Pl	2.55 (±0.41)	
Qz–Bt	2.59 (±0.46)	
Kfs-Bt	2.85 (±0.48)	
Rock		
Beishan granite (Liu et al. 2020)	1.18	CCNBD
Barre granite (Nasseri et al. 2010)	1.89	CCNBD
Newhurst granite (Whittaker et al. 1992)	1.72	HDB
Westerly granite (Zhang 2002)	2.61	CST
TGP granite (Yu 2001)	2.08	SENRBB

Note: CCNBD, chevron cracked notch Brazilian disc test; HDB, single edge cracked half-disc in three-point bending; CST, compact sample in tension; SENRBB, single edge notched round bar in bending

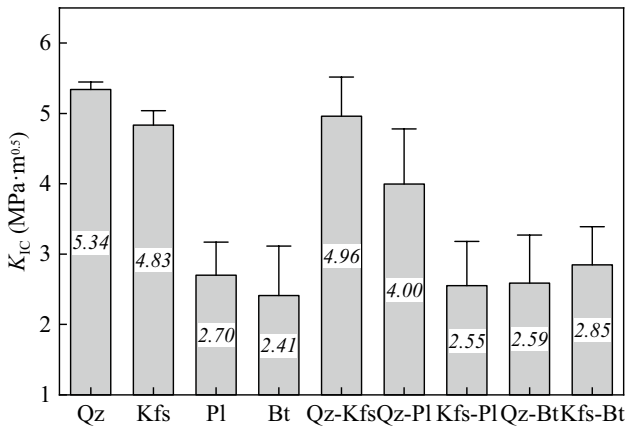


Fig. 12 Statistics of fracture toughness of minerals and mineral interfaces

U_{pp} can be calculated using the following mathematical relationship:

$$\frac{U_{pp}}{U_t} = 1 - \left[\frac{1 - 3\left(\frac{h_f}{h_{max}}\right)^2 + 2\left(\frac{h_f}{h_{max}}\right)^3}{1 - \left(\frac{h_f}{h_{max}}\right)^2}\right]. \tag{8}$$

Thus, the critical energy release rate G_c can then be calculated using the following:

$$G_c = \frac{\partial U_{crack}}{\partial A_c} = \frac{U_{crack}}{A_{max}}, \tag{9}$$

where the maximum contact area A_{max} is calculated by substituting h_{max} in Eq. (3).

Finally, Eq. (10) can be used to compute the tensile (mode **I**) fracture toughness K_{IC}

$$K_{IC} = \sqrt{G_c E_r}. \tag{10}$$

The fracture toughness of minerals and mineral interfaces in granite was calculated based on the aforementioned energy analysis method (Table 2). The statistical analysis of the obtained results (Fig. 12) demonstrates that the mean fracture toughness of quartz is the highest, thus reaching 5.34 (MPa·m$^{0.5}$), followed by K-feldspar, plagioclase, and biotite, with values of 4.83, 2.70, and 2.41 (MPa·m$^{0.5}$), respectively. For mineral interfaces, the fracture toughness of quartz–K-feldspar is the largest, followed by quartz–plagioclase, K-feldspar–biotite, quartz–biotite, and K-feldspar–plagioclase, with values of 4.96, 4.00, 2.85, 2.59, and 2.55 MPa·m$^{0.5}$, respectively.

The above-mentioned fracture toughness values of quartz, feldspar, and biotite are similar to the results reported by Mahabadi (2012), thus confirming the reliability of test results obtained from this study's meso-test methods. Furthermore, previous research on the fracture toughness of granite (Whittaker et al. 1992; Yu 2001; Zhang 2002; Nasseri et al. 2010; Liu et al. 2020) demonstrates that the macroscopic fracture toughness of granite is almost smaller than that of each phase in granite (Table 2). Indeed, this is related to the scale effects. Therefore, the scale effects of cracks initiated and propagated should be considered (Du et al. 2021; Kuruppu et al. 2014). The results show that additional experimental research is required to understand the effect of scaling on fracture toughness in heterogeneous materials such as granite.

Linear relationships between Young's moduli and fracture toughness of different granite minerals and mineral interfaces have been developed based on the data from the load–indentation depth curves (Table 3), which can be adopted to estimate the fracture toughness from Young's moduli at the mesoscale.

Figure 13 shows the relationships between Young's moduli and fracture toughness of K-feldspar and quartz–K-feldspar interface in granite. Based on the fitting formula, the fracture toughness of the minerals and mineral interfaces increases, because their Young's modulus increases. This is because the fracture strength is positively correlated with the square root of Young's modulus (Eq. (11)) (Lawn and Wilshaw 1975). An increase in Young's modulus can increase the ultimate fracture strength and improve a

Table 3 Fitting formula of minerals and mineral interfaces

Type	Fitting formula
Mineral	
Qz	$K_{IC} = 0.055E - 0.435$
Kfs	$K_{IC} = 0.060E - 1.218$
Pl	$K_{IC} = 0.032E + 0.561$
Bt	$K_{IC} = 0.072E - 1.017$
Mineral interface	
Qz–Kfs	$K_{IC} = 0.049E + 0.439$
Qz–Pl	$K_{IC} = 0.060E - 0.198$
Kfs–Pl	$K_{IC} = 0.048E - 0.207$
Qz–Bt	$K_{IC} = 0.068E - 1.227$
Kfs-Bt	$K_{IC} = 0.069E - 1.035$

material's ability to resist fracture (Liu et al. 2016; Yuan and Xi 2011)

$$\sigma_f = \sqrt{\frac{4E\gamma_p}{\pi l}}, \tag{11}$$

where σ_f, γ_p, and l are critical fracture stress, plastic energy, and crack length, respectively.

3.4 Fracture Characterization

The mesofracture patterns under multiple stress conditions must be considered to understand the mechanical properties of granite at the mesoscale. Therefore, μCT scanning of a failed Brazilian disc was performed to provide data on the mesofracture pattern for the subsequent calibration of macro- and mesomechanical parameters. A total of 2140 CT scans were resampled, and the resampled 3D model was longitudinally sliced at a spacing of 1 mm. Consequently, 12 CT gray images containing complete fracture cracks were obtained (Fig. 14a). Biotite, the densest mineral, shows the greatest brightness on CT images (white), followed by quartz (light gray) and feldspar (dark gray). To measure the length ratio, the crack traces shown in Fig. 14a are digitized and counted using the length analysis feature of the open-source software ImageJ. The ratio of intergranular (in addition to different mineral interfaces) cracks to intragranular (in different minerals) cracks was obtained by measuring the length of the fracture path. The mineral composition along the cracks was recorded. Note that ~ 84% of tensile cracks follow intragranular cracks and primarily occur in feldspar minerals, whereas ~ 16% follow grain boundaries and primarily occur at feldspar–biotite interfaces (Table 4). The statistics are consistent with result of Fujii et al. (2007), which states that ~ 90% of the tensile fractures follow the intragranular cracks and ~ 10% follow the grain boundaries observed using an optical microscopy. This indicates that multiple parts of the crack are intragranular, thus cutting across the rock-forming mineral grains.

The crack propagation path (Fig. 14b) demonstrates that crack propagation is highly dependent on the orientation of the mineral structure in granite, which is particularly prominent in biotite. The favorable orientation of the cleavage in the biotite or feldspar–biotite interface causes the branch of the crack and deviated cracks or the termination of the main crack because of the angle between the cleavage of biotite and the main crack. Locations (3)–(5) in Fig. 14b show the local termination of the primary crack when the cleavages

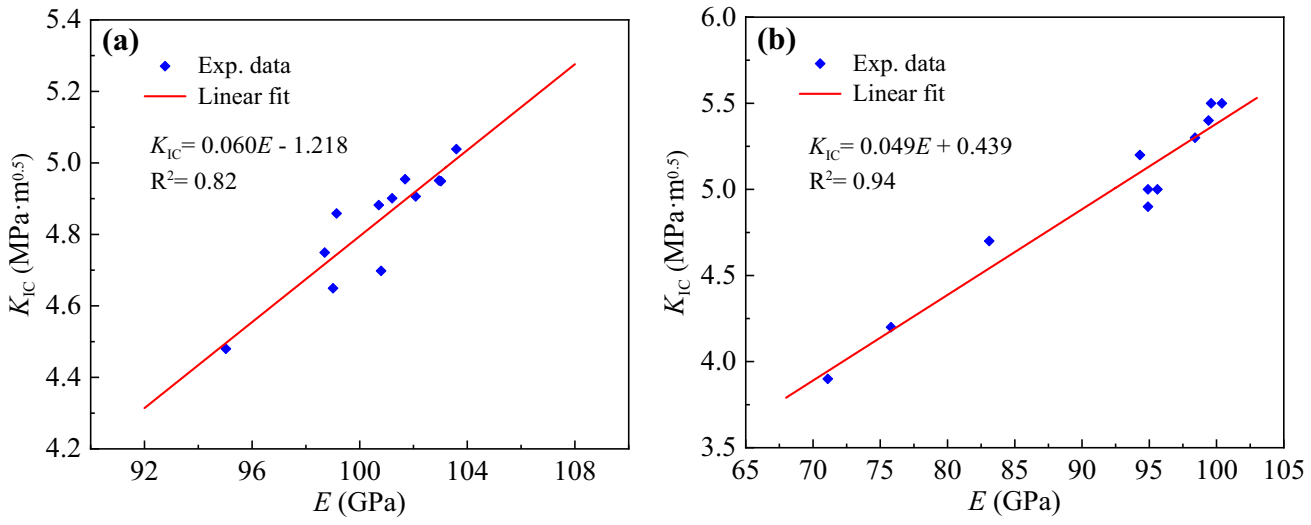


Fig. 13 Relationships between Young's moduli and fracture toughness of K-feldspar and quartz–K-feldspar interfaces in granite: **a** K-feldspar; **b** quartz–K-feldspar interface

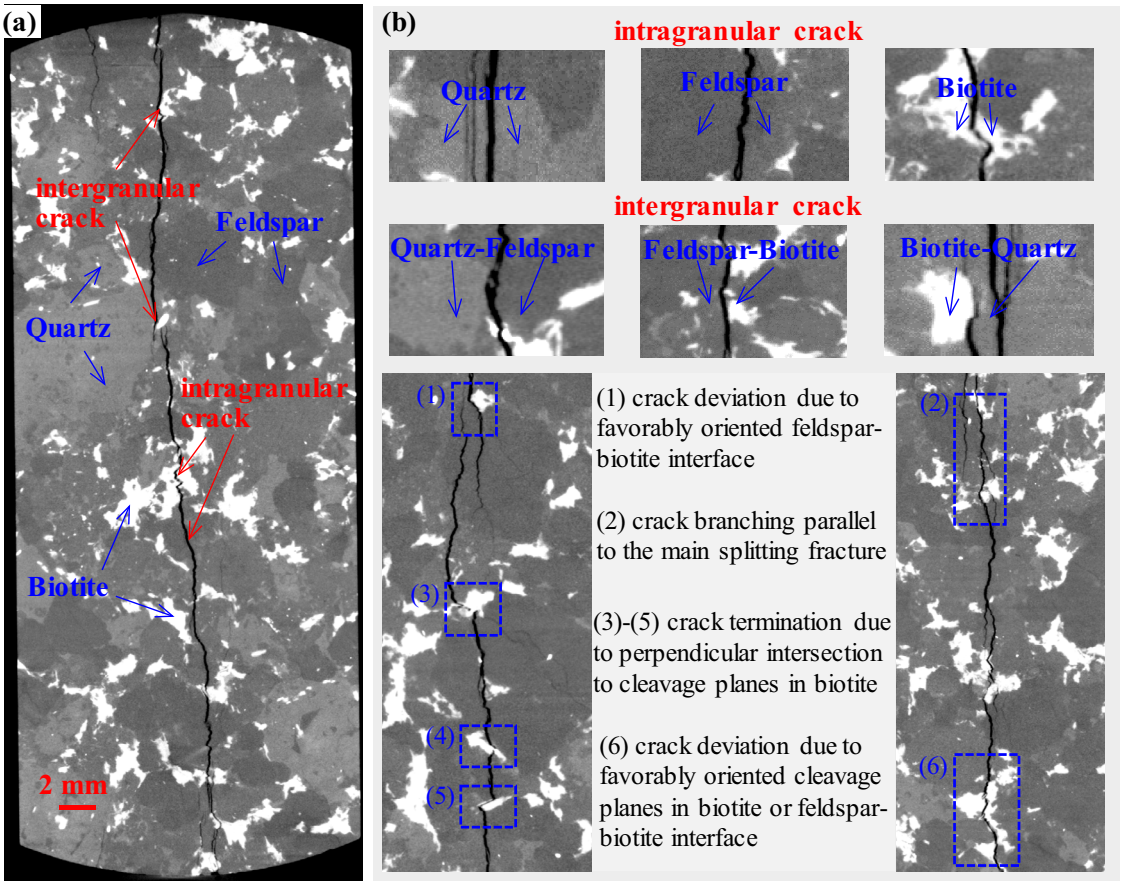


Fig. 14 Vertical μCT sliced images of failed Brazilian disc: **a** the entire crack observed in the μCT image; **b** typical failure patterns of the Brazilian disc observed in the μCT images

Table 4 Length of intragranular cracks in different minerals and intergranular cracks along with different mineral interfaces

Number	Length of cracks (mm)	Intragranular cracks (mm)			Intergranular cracks (mm)		
		Feld	Qz	Bt	Feld–Bt	Qz–Feld	Qz–Bt
1	67.5	34.5	17.3	3.6	3.8	3.2	5.0
2	76.6	36.7	25.2	4.0	3.7	2.6	4.3
3	79.7	43.0	20.9	6.8	3.3	2.8	3.0
4	69.1	32.9	14.0	11.7	3.5	4.3	2.6
5	65.4	31.8	12.1	7.4	3.9	3.2	7.0
6	71.7	38.0	10.9	11.1	5.7	5.1	1.0
7	81.4	43.2	16.6	6.0	7.1	3.3	5.2
8	76.4	33.2	16.8	15.6	3.4	3.0	4.5
9	70.3	40.4	15.3	4.0	5.2	4.0	1.4
10	65.8	33.2	18.9	3.6	4.9	2.5	2.7
11	76.2	35.9	21.6	5.8	6.0	3.9	3.0
12	66.8	28.5	18.6	9.0	5.0	4.8	1.0
Average	72.2	35.9	17.4	7.4	4.6	3.6	3.4
		60.7 (84%)			11.6 (16%)		

of biotite intersect the main crack at an unfavorable angle. In summary, the orientation of the biotite cleavage affects the crack's local propagation path. This observation is consistent with that of Mahabadi et al. (2014) and Nasseri et al. (2011) in evaluating the roughness of the fracture damage zone of granite.

4 Homogenization Upscaling Analysis

4.1 Upscaling Methods

Two homogenization methods, namely, generalized means (GM) and Mori–Tanaka (MT) methods, are used to upscale granite's Young's modulus from mesoscopic to macroscopic.

4.1.1 Generalized Means Method

The generalized means (GM) method is a semiempirical model for predicting the mechanical properties (e.g., Young's modulus E and bulk modulus K) of multiphase materials and polycrystalline rocks (Ji et al. 2004), which relates the effective properties of multiphase materials to their compositions. The mechanical properties of multiphase materials in the GM method are related to their volume fraction and composition, which can be determined using the following equations (Ayatollahi et al. 2020; Ji et al. 2004):

$$M_{\mathrm{d}}(J) = \left[\sum_{\mathrm{i}=1}^{\mathrm{n}} (f_{\mathrm{i}} M_{\mathrm{i}}^{J})\right]^{1/J} \tag{12}$$

$$\sum_{\mathrm{i}=1}^{\mathrm{n}} f_{\mathrm{i}} = 1, \tag{13}$$

where the subscripts i and d are the ith phase and multiphase materials comprising n phases, respectively; f_{i} is the volume fraction of each phase; M is a specific mechanical property; and J is a scaling parameter, which is called the mesostructural coefficient, because its value depends on the mesostructure.

Note that the J value of each multiphase material must be separately measured by performing multiple tests. In special cases, $M_{\mathrm{d}}(J=1)$, $M_{\mathrm{d}}(J=-1)$, $M_{\mathrm{d}}(J=\infty)$, and $M_{\mathrm{d}}(J=-\infty)$ yield the arithmetic mean, harmonic mean, maximum, and minimum values of M_{d}, respectively. For polycrystalline rocks (e.g., granite), the predicted Young's moduli at $J=-\infty$ are in good agreement with that from uniaxial compression tests (Ayatollahi et al. 2020).

4.1.2 Mori–Tanaka Method

The Mori–Tanaka (MT) method is popular in mesomechanics to estimate the homogenization of multiphase composites (Xu et al. 2020). This method assumes that inclusions are embedded in the matrix and interaction between different phases is considered. The average stress of the matrix interacts with the adjacent inclusions and the matrix, thus reflecting the interaction between inclusions (Mori and Tanaka 1973; Zhang et al. 2009).

The granite homogenization procedure based on the MT method is divided into two steps (Huang et al. 2014; Shen et al. 2013). The first step includes the creation of a homogeneous biotite porous matrix by homogenizing the pore structure of granite and solid phase of biotite particles (Eqs. (14) and (15)) (Huang et al. 2014). The second step includes the homogenization of quartz, K-feldspar, plagioclase, and biotite porous matrix into homogeneous materials equivalent to granite (Eqs. (16) and (17))

$$k_{\mathrm{low}} = \frac{4(1-\omega)k_{\mathrm{s}}\mu_{\mathrm{s}}}{4\mu_{\mathrm{s}} + 3\omega k_{\mathrm{s}}} \tag{14}$$

$$\mu_{\mathrm{low}} = \frac{(1-\omega)\mu_{\mathrm{s}}}{1 + 6\omega(k_{\mathrm{s}} + 2\mu_{\mathrm{s}})/(9k_{\mathrm{s}} + 8\mu_{\mathrm{s}})} \tag{15}$$

$$K_{\mathrm{hom}} = \left(\sum_{\mathrm{r}=1}^{3} \frac{f_{\mathrm{r}} k_{\mathrm{r}}}{3k_{\mathrm{r}} + 4\mu_{\mathrm{low}}}\right)\left(\sum_{\mathrm{r}=1}^{3} \frac{f_{\mathrm{r}}}{3k_{\mathrm{r}} + 4\mu_{\mathrm{low}}}\right)^{-1} \tag{16}$$

$$G_{\mathrm{hom}} = \frac{\sum_{\mathrm{r}=1}^{3} \frac{f_{\mathrm{r}}\mu_{\mathrm{r}}}{\mu_{\mathrm{low}}(9k_{\mathrm{low}}+8\mu_{\mathrm{low}})+6\mu_{\mathrm{r}}(k_{\mathrm{low}}+2\mu_{\mathrm{low}})}}{\sum_{\mathrm{r}=1}^{3} \frac{f_{\mathrm{r}}}{\mu_{\mathrm{low}}(9k_{\mathrm{low}}+8\mu_{\mathrm{low}})+6\mu_{\mathrm{r}}(k_{\mathrm{low}}+2\mu_{\mathrm{low}})}}, \tag{17}$$

where k_{s}, k_{low}, μ_{s}, and μ_{low} are the bulk and shear moduli of the biotite and biotite porous matrix, respectively; ω is the porosity of the biotite pore matrix; K_{hom} and G_{hom} are the bulk and shear moduli of the granite, respectively; f_{r}, k_{r}, and μ_{r} are the volume fraction, bulk modulus, and shear modulus of minerals, respectively; and r = 1, 2, 3 are quartz, K-feldspar, and plagioclase, respectively. The bulk and shear moduli of each mineral are obtained by $k = E/3(1-2\nu)$ and $\mu = E/2(1+\nu)$.

Finally, Eq. (18) can be used to compute the homogenization of granite's Young's modulus E_{hom}

$$E_{\mathrm{hom}} = \frac{9K_{\mathrm{hom}}G_{\mathrm{hom}}}{3K_{\mathrm{hom}} + G_{\mathrm{hom}}}. \tag{18}$$

4.2 Comparison Between Homogenization Results and Uniaxial Compression Test Results

Table 5 and Fig. 15 compare the Young's moduli of granite obtained from the uniaxial compression test with those obtained using the two upscaling methods. To summarize the rules better, the experimental data of six related granites from the previous literature are provided alongside the calculated results of this study. The error percentages demonstrate that the GM method can reasonably predict granite's Young's modulus. The error percentage of other prediction results was < 10%, except for two instances (reference II and

Table 5 Young's moduli for some minerals and granite rocks

Reference	Mineral	Young's moduli of minerals (GPa)	Volume fraction (%)	Young's modulus of granite (GPa)			Error 1 (%)	Error 2 (%)
				GM method	MT method	Uniaxial compression test		
I: Zhang et al. (2017)	Quartz	86.1	0.09	30.6*	36.1	31.0	1.2	16.5
	Feldspar	35.9	0.66					
	Mica	30.2	0.25					
II: Ma et al. (2021); Blake et al. 2019	Quartz	97.4	0.58	40.7*	78.6*	52.1	21.9	50.7
	Feldspar	64.2	0.38					
	Mica	39.3	0.04					
III: Mahabadi (2012)	Quartz	99.3	0.22	38.2*	76.0*	52.0	26.5	46.2
	Feldspar	75.6	0.73					
	Biotite	37.1	0.05					
IV	Quartz	106.3	0.40	45.2	76.3	41.4	9.1	84.3
	K-feldspar	100.6	0.07					
	Plagioclase	67.2	0.40					
	Biotite	45.2	0.13					
V: Goldsmith et al. (1976); Ayatollahi et al. (2020)	Quartz	106.7	0.319	44.8	77.0*	46.1	2.8	67.1
	Microcline	76.7	0.178					
	Albite	69.9	0.365					
	Biotite	44.2	0.108					
	Muscovite	79.3	0.03					
VI: Duevel and Haimson (1997); Ayatollahi et al. (2020)	Quartz	106.7	0.3	70.6	83.4*	74.2	4.8	12.3
	Microcline	76.7	0.4					
	Albite	69.9	0.2					
	Muscovite	79.3	0.1					
VII: Swan (1978)	Quartz	95.7	0.44	70.3*	88.3*	69.4	1.4	27.2
	K-feldspar	73.9	0.12					
	Plagioclase	88.1	0.39					
	Biotite	68.3	0.05					

Note: "Error 1" is the error between the GM method and uniaxial compression test results; "Error 2" is the error between the MT method and uniaxial compression test results; the number with "*" is calculated by the authors using these two methods

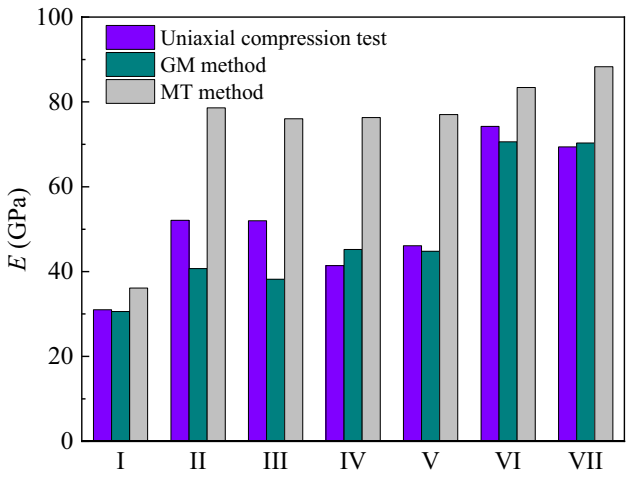


Fig. 15 Comparison between the predicted and experimental moduli data from Table 5

III). However, the MT method cannot reasonably predict the Young's modulus of granite, and its prediction results overestimate the Young's modulus of granite by 12.3–84.3%.

The failure of the MT method's prediction may be attributed to its two major limitations: (1) The method ignores the effects of the size and number of inclusions and assumes that the shape of the inclusion is ellipsoid, and (2) the method applies only to composites whose inclusions have volume fractions that are substantially lower than that of the matrix, and the matrix is usually the phase with the weakest mechanical properties. As shown in Table 5, the weakest phase (primarily biotite) does not have the highest volume fraction, and inclusions such as feldspar have a significantly high volume fraction to be considered a "low-fraction inclusion" in the MT method. Literature (Guéry et al. 2010; Luo et al. 2020; Zhang et al. 2018) demonstrates that the MT method can accurately predict claystone and shale that contain a dominant percentage of clay matrices. However, the Young's modulus of polycrystalline

rocks, such as granite, cannot be accurately predicted. The GM method can offer a new alternative for predicting the Young's modulus of polycrystalline rocks.

5 Conclusions

This study examined the mechanical properties and fracture characteristics of granite at the mesoscale using nanoindentation and μCT. The Young's modulus of granite was calculated using the homogenization upscaling method that was then compared with the uniaxial compression test results. The main conclusions are summarized as follows:

(1) The mechanical properties of granite minerals are related to the peak load. Young's modulus and hardness of quartz, K-feldspar, and plagioclase decrease as the peak load increases and tend to stabilize when the peak load reaches 5000 μN, whereas the Young's moduli and hardness of biotite at various peak loads are in variability and irregularity.
(2) Under Brazilian splitting conditions, ~ 84% of the tensile cracks along intragranular cracks primarily occur in feldspar minerals, while ~ 16% along grain boundaries mainly occur at feldspar–biotite interfaces. The crack propagation in granite is highly dependent on the orientation of the mineral structure, particularly in biotite.
(3) The fracture toughness and Young's modulus of granite minerals have a linear relationship. The Young's modulus of granite increases with the weakest mineral's Young's modulus and decreases with the weakest mineral's volume fraction. The GM method is more suitable for predicting the Young's modulus of granite than the MT method.

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Data Availability Datasets related to this article are available from the corresponding author (dpxu@whrsm.ac.cn).

Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Akono AT, Kabir P, Shi ZF et al (2019) Modeling CO_2–induced alterations in Mt. Simon sandstone via nanomechanics. Rock Mech Rock Eng 52:1353–1375

Ayatollahi MR, Najafabadi MZ, Koloor SSR et al (2020) Mechanical characterization of heterogeneous polycrystalline rocks using nanoindentation method in combination with generalized means method. J Mech 36:813–823

Blake OO, Faulkner DR, Tatham DJ (2019) The role of fractures, effective pressure and loading on the difference between the static and dynamic Poisson's ratio and Young's modulus of Westerly granite. Int J Rock Mech Min Sci 116:87–98

Chen X, Ogasawara N, Zhao MH et al (2007) On the uniqueness of measuring elastoplastic properties from indentation: the indistinguishable mystical materials. J Mech Phys Solids 55(8):1618–1660

Chen Y, Hou YQ, Ji XP et al (2021) Characterization of surface mechanical properties of various aggregates from micro scale using AFM. Constr Build Mater 286:122847

Cheng YT, Li ZY, Cheng CM (2002) Scaling relationships for indentation measurements. Philos Mag A 82:1821–1829

Constantinides G, Chandran R-KS, Ulm FJ et al (2006) Grid indentation analysis of composite microstructure and mechanics: principles and validation. Mater Sci Eng A 430:189–202

Domnich V, Gogotsi Y, Dub S (2000) Effect of phase transformations on the shape of the unloading curve in the nanoindentation of silicon. Appl Phys Lett 76:2214–2216

Du JT, Whittle A-J, Hu LM et al (2021) Characterization of meso–scale mechanical properties of Longmaxi shale using grid microindentation experiments. J Rock Mech Geotech Eng 13:555–567

Duevel B, Haimson B (1997) Mechanical characterization of pink Lac du Bonnet granite: evidence of nonlinearity and anisotropy. Int J Rock Mech Min Sci 34:543–543

Eliyahu M, Emmanuel S, Ruarri J et al (2015) Mechanical properties of organic matter in shales mapped at the nanometer scale. Mar Pet Geol 59:294–304

Fujii Y, Takemura T, Takahashi M et al (2007) Surface features of uniaxial tensile fractures and their relation to rock anisotropy in Inada granite. Int J Rock Mech Min Sci 44:98–107

Goldsmith W, Sackman J-L, Ewerts C (1976) Static and dynamic fracture strength of Barre granite. Int J Rock Mech Min Sci 13:303–309

Guéry AAC, Cormery F, Shao JF, Kondo D (2010) A comparative micromechanical analysis of the effective properties of a geomaterial: effect of mineralogical compositions. Comput Geotech 37(5):585–593

Hay J (2009) Introduction to instrumented indentation testing. Exp Tech 33(6):66–72

Hu CL, Li ZJ (2015) A review on the mechanical properties of cement–based materials measured by nanoindentation. Constr Build Mater 90:80–90

Huang Y, Shen WQ, Shao JF et al (2014) Multi–scale modeling of time-dependent behavior of claystones with a viscoplastic compressible porous matrix. Mech Mater 79:25–34

Ji SC, Wang Q, Xia B et al (2004) Mechanical properties of multiphase materials and rocks: a phenomenological approach using generalized means. J Struct Geol 26:1377–1390

Jiang Q, Yang B, Yan F et al (2021) Morphological features and fractography analysis for in situ spalling in the China Jinping underground laboratory with a 2400 m burial depth. Tunn Undergr Space Technol 118:104194

Juliano T, Gogotsi Y, Domnich V (2003) Effect of indentation unloading conditions on phase transformation induced events in silicon. J Mater Res 18:1192–1201

Kuruppu MD, Obara Y, Ayatollahi MR et al (2014) ISRM–suggested method for determining the mode I static fracture toughness using semi–circular bend specimen. Rock Mech Rock Eng 47:267–274

Lawn BR, Wilshaw TR (1975) Fracture of brittle solids. Cambridge University Press, London

Li Y, Yang JH, Pan ZJ et al (2020) Nanoscale pore structure and mechanical property analysis of coal: an insight combining AFM and SEM images. Fuel 260:116352
Li YC, Luo SM, Lu M et al (2021) Cross–scale characterization of sandstones via statistical nanoindentation: evaluation of data analytics and upscaling models. Int J Rock Mech Min Sci 142:104738
Liu KQ, Ostadhassan M (2017) Microstructural and geomechanical analysis of Bakken shale at nanoscale. J Pet Sci Eng 153:133–144
Liu KQ, Ostadhassan M, Bubach B (2016) Applications of nano–indentation methods to estimate nanoscale mechanical properties of shale reservoir rocks. J Nat Gas Sci Eng 35:1310–1319
Liu L, Du GY, Li EB et al (2020) Fracture mechanical behaviors and acoustic emission characteristics of Beishan granite under CCNBD test. Chin J Rock Mech Eng 39(S2):3237–3244
Luo SM, Lu YH, Wu YK et al (2020) Cross–scale characterization of the elasticity of shales: statistical nanoindentation and data analytics. J Mech Phys Solids 140:103945
Luo SM, Kim DY, Wu YK et al (2021) Big data nanoindentation and analytics reveal the multi-staged, progressively-homogenized, depth-dependent upscaling of rocks' properties. Rock Mech Rock Eng 54:1501–1532
Ma ZY, Gamage RP, Zhang CP (2020) Application of nanoindentation technology in rocks: a review. Geomech Geophys Geo 6(4):1–27
Ma ZY, Zhang CP, Gamage RP et al (2021) Uncovering the creep deformation mechanism of rock-forming minerals using nanoindentation. Int J Min Sci Technol 32(2):283–294
Mahabadi OK (2012) Investigating the influence of micro–scale heterogeneity and microstructure on the failure and mechanical behaviour of geomaterials. Toronto University, Canada
Mahabadi OK, Tatone BSA, Grasselli G (2014) Influence of microscale heterogeneity and microstructure on the tensile behavior of crystalline rocks. J Geophys Res-Solid Earth 119:5324–5341
Manjunath GL, Jha B (2019) Nanoscale fracture mechanics of Gondwana coal. Int J Coal Geol 204:102–112
Maruvanchery V, Kim E (2020) Mechanical characterization of thermally treated calcite–cemented sandstone using nanoindentation, scanning electron microscopy and automated mineralogy. Int J Rock Mech Min Sci 125:10415
Mavko G, Mukerji T, Dvorkin J (2009) The rock physics handbook. Cambridge University Press, London
Miller M, Bobko C, Vandamme M, Ulm FJ (2008) Surface roughness criteria for cement paste nanoindentation. Cem Concr Res 38:467–476
Mori T, Tanaka K (1973) Average stress in matrix and average elastic energy of materials with misfitting inclusions. Acta Metall 21:571–574
Nasseri MHB, Grasselli G, Mohanty B (2010) Fracture toughness and fracture roughness in anisotropic granitic rocks. Rock Mech Rock Eng 43:403–415
Nasseri MHB, Rezanezhad F, Young RP (2011) Analysis of fracture damage zone in anisotropic granitic rock using 3D X-ray CT scanning techniques. Int J Fract 168:1–13
Oliver WC, Pharr GM (1992) An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments. J Mater Res 7:1564–1583
Pöhl F (2019) Pop–in behavior and elastic–to–plastic transition of polycrystalline pure iron during sharp nanoindentation. Sci Rep 9:1564–1583
Shen WQ, Kondo D, Dormieux L et al (2013) A closed–form three scale model for ductile rocks with a plastically compressible porous matrix. Mech Mater 59:73–86
Shi X, Jiang S, Lu SF et al (2019) Investigation of mechanical properties of bedded shale by nanoindentation tests: a case study on Lower Silurian Longmaxi Formation of Youyang area in southeast Chongqing, China. Petrol Explor Dev 46:163–172
Sobhbidari F, Hu QH (2021) Recent advances in the mechanical characterization of shales at nano-to micro-scales: a review. Mech Mater 162:104043
Sun CL, Li GC, Gomah M-E et al (2020) Meso–scale mechanical properties of mudstone investigated by nanoindentation. Eng Fract Mech 238:107245
Sun CL, Li GC, Xu JH et al (2021) Rheological characteristics of mineral components in sandstone based on nanoindentation. Chin J Rock Mech Eng 40(1):77–87
Swan (1978) The mechanical properties of Stripa grantite. Lawrence Berkeley National Laboratory
Tabor D (1978) Phase transitions and indentation hardness of Ge and diamond. Nature 273:406–406
Tyurin AI, Victorov SD, Kochanov AN et al (2016) Methods of micro- and nanoindentation for characterization of local physical and mechanical properties of multiphase materials. AIP Conf Proc 1783:020227
Whittaker BN, Singh RN, Sun G (1992) Rock fracture mechanics: principles, design, and applications. Elsevier
Wu YK, Li YC, Luo SM et al (2020) Multiscale elastic anisotropy of a shale characterized by cross-scale big data nanoindentation. Int J Rock Mech Min Sci 134:104458
Xie HP, Li CB, Gao MZ et al (2021) Conceptualization and preliminary research on deep in situ rock mechanics. Chin J Rock Mech Eng 40(02):217–232
Xu JJ, Tang XH, Wang ZZ et al (2020) Investigating the softening of weak interlayers during landslides using nanoindentation experiments and simulations. Eng Geol 277:105801
Yang C, Xiong YQ, Wang J-F et al (2020) Mechanical characterization of shale matrix minerals using phase–positioned nanoindentation and nano–dynamic mechanical analysis. Int J Coal Geol 229:103571
Yu (2001) Measuring properties of rock from the site of permanent shiplock in three Gorges project. Test Report, Yangtze River Scientific Research Institute
Yuan CC, Xi XK (2011) On the correlation of Young's modulus and the fracture strength of metallic glasses. J Appl Phys 109:033515
Zhang ZX (2002) An empirical relation between mode I fracture toughness and the tensile strength of rock. Int J Rock Mech Min Sci 39:401–406
Zhang G, Wei Z, Ferrell RE (2009) Elastic modulus and hardness of muscovite and rectorite determined by nanoindentation. Appl Clay Sci 43:271–281
Zhang F, Guo HQ, Zhao JJ et al (2017) Experimental study of micro–mechanical properties of granite. Chin J Rock Mech Eng 36:3864–3872
Zhang F, Guo HQ, Hu DW et al (2018) Characterization of the mechanical properties of a claystone by nano–indentation and homogenization. Acta Geotech 13:1395–1404

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